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Energy-Efficient Architecture and Dataflow Optimization for Spiking Neural Network Accelerators

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Doctor of Philosophy
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To my parents, family and friends for their great support

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Publications

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Abstract

Energy-Efficient Architecture and Dataflow Optimization for Spiking Neural Network Accelerators

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Spiking neural networks (SNNs) offer a promising biologically-plausible computing model and lend themselves to ultra-low-power event-driven processing on neuromorphic processors. Compared with the conventional artificial neural networks, SNNs are well-suited for processing complex spatiotemporal data. In this dissertation, we aim to address key difficulties in accelerating SNNs: developing bio-plausible and hardware friendly algorithm, efficient processing of the added temporal dimension, and handling unstructured sparsity emergent in both space and time.

First, training SNNs to reach the same performances of conventional deep artificial neural networks (ANNs), particularly with error backpropagation (BP) algorithms, poses a significant challenge due to inherent complex dynamics and non-differentiable spike activities of spiking neurons. In this dissertation, we present the first study on realizing competitive spike-train level backpropagation (BP) like algorithms to enable on-chip training of SNNs. We propose a novel spike-train level direct feedback alignment (ST-DFA) algorithm, which is much more bio-plausible and hardware friendly than BP. Algorithm and hardware co-optimization and efficient online neural signal computation are explored for on-chip implementation of ST-DFA. On the Xilinx ZC706 FPGA board, the proposed hardware-efficient ST-DFA shows excellent performance vs. overhead trade-offs for real-world speech and image classification applications.

Despite its significance, dataflow optimization of spiking neural accelerator architec-

tures has not been extensively studied. Recognizing the need for efficient processing of complex spatiotemporal data while considering the all-or-none nature of spiking activities, we propose holistic reconfigurable dataflow optimization for systolic array acceleration of spiking convolutional networks (S-CNNs). A novel scheme is introduced for parallel acceleration of computation across multiple time points, which further allows for systemic optimization of variable tiling for a large performance and efficiency gains. Also, we pack multiple time points into a single time window (TW) and process the computations induced by active synaptic inputs falling under several TWs in parallel, leading to the proposed parallel time batching. It allows weight reuse across multiple time points and enhances the utilization of the systolic array with reduced idling of processing elements, overcoming the irregularity of sparse firing activities. We optimize the granularity of time-domain processing, i.e., the TW size, which significantly impacts the data reuse and utilization.

Lastly, we propose a novel technique and architecture that allow the exploitation of temporal information compression with structured sparsity and parallelism across time, and significantly improves data movement on a systolic array. We split a full range of temporal domain into several time windows (TWs) where a TW packs multiple time points, and encode the temporal information in each TW with Split-Time Temporal coding (STT) by limiting the number of spikes within a TW up to one. STT enables sparsification and structurization of irregular firing activities and dramatically reduces computational overhead while delivering competitive classification accuracy without a huge drop. To further improve the data reuse, we propose an Integration Through Time (ITT) technique that processes integration steps across different TWs in parallel with a systolic array.

In this dissertation, we provide unique and powerful solutions for the efficient acceleration of the spiking models with various datasets.

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Chapter 1

Introduction

1.1 Neuromorphic Computing Systems

Conventional non-spiking artificial neural network models, or simply ANNs, employ only rate coding where continuous-valued signals resulted from activation functions such as sigmoid and rectified linear unit (ReLU) correspond to average firing rates. On the other hand, spiking neural networks (SNNs) more closely resemble biological neurons, explicitly model all-or-none firing spikes across both space and time, and can leverage a rich family of rate and temporal codes for complex spatiotemporal information processing.

Recent studies reported competitive performances for various image and speech tasks with biologically inspired [1, 2] and backpropagation based [3, 4] SNN training methods. With great potentials in ultra-low power event-driven learning leveraging the spatiotemproal dynamics of SNNs [5], neuromorphic processors have gathered significant interest in both academia and industry, resulting in well-known neuromorphic chips including IBM's TrueNorth [6] and Intel's Loihi [7].

1.2 Spiking Neural Network Algorithms

Despite the recent progresses in SNNs and neuromorphic processor designs, fully leveraging the theoretical computing advantages of SNNs over traditional artificial neural networks (ANNs) [5] to achieve the state-of-the-art performance for real-world applications remains challenging. One chief difficulty here lies in training of SNNs in terms of achievable performance and computational complexity.

Inspired by the success of error backpropagation (BP) and its variants such as stochastic gradient decent in training conventional ANNs [8], various SNN BP methods have emerged, aiming at attaining the same level of performance [9, 10, 11, 4]. The major challenges in BP training of SNNs stem from the non-differentiablity of spike events and temporal dynamics that prevent straightforward derivative computation. SpikeProp [9] is the first BP algorithm to train SNNs by BP. However, SpikeProp is limited to single-spike training for learning simple functions like XOR. [10] proposes a BP algorithm which differentiates neuron's membrane potential instead of discrete output spikes. [11] improves [10] by capturing temporal effects with backpropapogation through time (BPTT) [12]. However, the error gradient is still computed by differentiating the membrane potential, leading to inconsistency w.r.t the rate-coded loss function. More recently, [4] proposes a hybrid macro/micro level backpropagation (HM2-BP) algorithm for training multi-layer

SNNs, which addresses the aforementioned issues. HM2-BP precisely captures the temporal behavior of the SNN at the microscopic level and directly computes the gradient of the rate-coded loss function w.r.t tunable parameters. As a result, HM2-BP demonstrates the state-of-the art learning performances on widely adopted SNN benchmarks such as MNIST [13] and Neuromorphic-MNIST (N-MNIST) [14], outperforming all other existing BP algorithms based on the leaky integrate-and-fire model.

While achieving excellent results, the aforementioned SNN BP algorithms are hampered by several limitations. The error signal is propagated backward layer by layer through weights symmetric to the feed-forward weights. This is considered not biologically-plausible. Furthermore, BP algorithms involve complex layer-by-layer backward computations, which is expensive to implement on-chip and introduces high training latency. For instance, while HM2-BP improves the scalability of BPTT [11] by operating on the spike-train level, i.e. application of BP does not discretize time, it still involves complex computations and its latency in the backward phase is proportional to network depth.

1.3 Spiking Neural Network Accelerators

Spiking neural networks (SNNs) more closely resemble biological neurons, explicitly model all-or-none firing spikes across both space and time, and can leverage a rich family of rate and temporal codes for complex spatiotemporal information processing. Recent studies reported competitive performances for various image and speech tasks with biologically inspired [1, 2] and backpropagation based [3, 4] SNN training methods. With great potentials in ultra-low power event-driven learning leveraging the spatiotemproal dynamics of SNNs [5], neuromorphic processors have gathered significant interest in both academia and industry, resulting in well-known neuromorphic chips including IBM's TrueNorth [6] and Intel's Loihi [7].

Nevertheless, hardware acceleration of spike-based models is complicated by temporal computation and sparse spiking activities in both space and time, two new challenges that are absent in accelerators of non-spiking networks such as DNNs. The added temporal dimension is fundamental to SNNs but introduces difficulties in managing compute and data movement. Furthermore, biological brains and engineered SNN models often exhibit a great deal of firing activity sparsity across both space and time, manifesting their promising efficiency. The sparse spiking activities of a well-trained SNN may vary from neurons to neurons, and from time points to time points. To fully explore the benefits of SNNs, one must address the challenges brought by irregular patterns of spatial and temporal sparsity.

Compared to the large body of work on DNN accelerators, e.g., [15, 16, 17, 18, 19], much less research has been devoted to SNN hardware accelerator architectures [20, 21, 22, 23, 24]. The two best-known industrial neuromorphic chips, IBM's TrueNorth [6] and Intel's Loihi [7], are based on a many-core architecture, comprising neuro-synaptic cores with an asynchronous mesh for core-to-core communication. Each neuromorphic core emulates a certain number of spiking neurons in a time-sequential manner. While both architectures target large-scale spiking neural computations with low power, there exist two primary disadvantages in these two designs: 1) lack of parallelism in each core: the computations associated with different spiking neurons are executed sequentially, one neuron at a time, and from time points to time points; and 2) assumption of large core memory: as opposed to many practical cases, it is assumed that all weights of the network are fully stored on-chip, and hence efficient dataflows maximizing the reuse of weight data are not targeted. These issues limit the achievable throughput and/or do not well support SNN acceleration on resource-constrained hardware like ones for edge computing.

The recent SNN architecture SpinalFlow explores a novel compressed, time-stamped, and sorted spike input/output representation [20]. The main drawback of SpinalFlow is that it only targets the class of temporally-coded spiking neuronal models in which each neuron fires at most once, which is a highly restrictive type and has limited accuracy for challenging learning tasks [25, 26]. While the smart exploration of such extreme temporal sparsity leads to large latency and energy efficiency benefits, SpinalFlow is not applicable to broader classes of SNNs employing rate and other types of temporal codes or a combination of thereof for high-accuracy decision making. Since the maximum firing count for each neuron is one, the structured sparse firing activities are handled as chronologically sorted inputs with a dearth of parallel acceleration through time.

1.4 Outline

The summary of the rest of this dissertation is as follows: In Chapter 2, we provide a brief background on unique characteristics of SNNs, basic operations in spiking neuron models, datasets and systolic arrays. In Chapter 3, we perform algorithm-hardware co-optimization and demonstrate the first on-chip hardware realization of Spike-Train level Direct Feedback Alignment (ST-DFA) for SNNs with significantly improved accelerator performance compared to conventional Back-Propagation (BP). In Chapter 4, we propose holistic reconfigurable dataflow optimization for systolic array acceleration of spiking convolutional networks (S-CNNs), and introduce SNN dataflow simulator to support systemic design space exploration. In Chapter 5, we introduce a novel technique for parallel acceleration in both space and time based on simultaneous processing of multiple time batches with a temporal granularity defined by the Time Window (TW) size, along with Spatiotemporally-non-overlapping Spiking Activity Packing (StSAP). In Chapter 6, we explore the proposed parallel acceleration technique to decouple the processing of feedforward synaptic connections from that of recurrent connections, for efficient acceler-